Daniel Thomas

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.62
St. Thomas Central School	Class XII (CBSE)	2014	89.20%
St. Thomas Central School	Class X (CBSE)	2012	10.00%

WORK EXPERIENCE

AI-Driven Autonomous Vehicle Systems, TiHAN

- **Developed an autonomous driving stack** integrating LiDAR/GNSS/CAMERA, ROS-based DBW, perception, localization, and control; validated on vehicles up to **80 km/h**.
- **Built LiDAR-based AI perception and localization** for 3D object tracking, velocity estimation (**<0.15 m/s RMSD**), and optimized NDT localization (**<10 ms**) with GPS-denied relocalization.
- **Developed LPV-NMPC, ACC, and AEB** for real-time trajectory tracking; demonstrated autonomous indoor shuttle navigation for DMRC**,** enabling reliable point-to-point transit without GNSS.

Senior Software Engineer (ML/AI), QUEST Global

- **Built an enterprise RAG system from scratch** over **10K+ pages** of technical documentation, enabling semantic retrieval and LLM-based generation for domain-specific engineering knowledge.
- **Built an LLM + RAG-based test-case generation pipeline for Ferrari**, automatically retrieving feature-specific dashboard documentation and generating **6-8 functional test cases per feature**, transforming manual documentation analysis into an automated AI workflow.
- **Developed semantic similarity search** across **7K+ JIRA issues** using embeddings and vector search, enabling engineers to discover historical issues and solutions before creating new tickets; deployed as a Chrome extension.
- **Engineered computer vision and automation pipelines** across **200K+ images**, reducing redundant training data by **30%** and achieving **93% pothole-detection accuracy.**

AI/ML Engineer, Tata Consultancy Services

- **Developed AI/ML and Computer Vision solutions** for autonomous vehicles for a major German-based OEM.
- **Engineered multi-class object detection** for automated anonymization of faces, license plates, location signs, and sensitive information across varying scales and orientations.
- **Optimized an image similarity-search platform to <30 ms retrieval latency**, deploying embedding-based search through Dockerized APIs; also developed real-time vehicle/pedestrian detection and tracking using camera and radar data.

PROJECTS

GNSS-DENIED NAVIGATION & EDGE AI, TiHAN

- **Led a 4-member team** to build a Jetson Orin-based **autonomous drone**, integrating Deep Learning, Computer Vision, Edge AI, sensor fusion, 5G, and cloud **across 20+ missions.**
- **Built GNSS-denied visual localization** using LightGlue deep-learning feature matching against satellite imagery, achieving **<5 m positioning accuracy at ~3 FPS**.
- **Designed 5G edge-to-cloud** command-and-control platform using MQTT-over-TLS, WebSockets, telemetry/state management, and reliable command acknowledgment, enabling remote mission planning, monitoring, and control.

Resume-to-Jobs Agent

- Built and deployed an **agentic job-matching system** with LangGraph and a custom MCP server spanning 6 job sources, supporting local Ollama/Gemini models, a live Streamlit UI, and Docker Compose deployment.
- Implemented **deterministic grounding and prompt-injection defenses**, removing ~27% of ungrounded LLM skill claims and keeping **0 adversarial postings in the top 5** across a 45-job evaluation.
- Built end-to-end **OpenTelemetry tracing and offline evaluation** with nDCG and a calibrated LLM judge, catching a ranking regression before release.

AI-Assisted Clinical Triage System

- Architected a **hybrid AI triage system** combining **LLM-based symptom extraction** with deterministic **medical protocols** for reproducible, auditable clinical decisions.
- Built a **multimodal intake pipeline** integrating **voice, vision, and free-text inputs** into a structured, human-verified patient state, with **Web Speech API + offline Vosk** fallback.
- Delivered an **end-to-end Python prototype** with patient kiosk, nurse dashboard, and AI pipeline.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL

Frameworks & Tools: PyTorch, Scikit-Learn, OpenCV, LangChain, LangGraph, Docker, Git, Linux, REST APIs

AI/ML: Machine Learning, Deep Learning, Generative AI, LLMs, RAG, Computer Vision, Agentic AI, Multimodal AI, Semantic Search, Vector Databases

Robotics & Autonomous Systems: ROS, Autonomous Navigation, Sensor Fusion and Control Algorithms

PATENTS AND PUBLICATIONS

Indian Patent Filed, 202641107246: GNSS-Denied Unmanned Aerial Vehicle Navigation System And Method Therefor.

Daniel Thomas et al., "**Real-Time Nonlinear Model Predictive Contouring Control for Autonomous Navigation in Complex Urban Geometries**" accepted in **IEEE ICRAS 2026, Japan.**